Sk Abu Tahir

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LinkedIn | GitHub | Portfolio

Education

Jalpaiguri Government Engineering College — West Bengal, India
B.Tech in Information Technology — *Expected 2028*

- **SGPA:** 9.48/10
- **Core Coursework:** Data Structures, Algorithms, Object-Oriented Programming, System Design

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C, C++

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend & Databases: Node.js, Express.js, Prisma, PostgreSQL, MongoDB, MySQL, Supabase

Core Concepts & Architecture: System Design, Object-Oriented Programming (OOP), RESTful APIs, RBAC

AI & Machine Learning: TensorFlow, PyTorch, Deep Learning (CNN, LSTM), Computer Vision, Scikit-learn, Pandas

Tools & Cloud: Git, GitHub, Linux, Postman, Vercel, Aiven, Railway

Experience

National Institute of Technology Jamshedpur — Jamshedpur, Jharkhand (Hybrid)
Research Intern — *Jun 2026 – Jul 2026*

- Engineered a hybrid CNN-LSTM image captioning model using TensorFlow on the Flickr8k dataset.
- Achieved a BLEU-1 score of 0.55, optimizing model performance for accurate natural language descriptions.
- Conducted research under the guidance of Dr. Mayukh Sarkar, applying advanced Deep Learning techniques.

DeepDuo Foundation — Kolkata, West Bengal
Project Intern — *Feb 2026 – May 2026*

- Architected a conference management platform using Next.js and Node.js supporting 50+ users.
- Developed scalable APIs, submission workflows, review pipelines, and secure authentication systems.
- Delivered 5+ production-ready features while collaborating in an agile team environment.

International Institute of SDGs Public Policy Research (IISPPR) — Remote
Full-Stack Developer Intern — *Aug 2025 – Dec 2025*

- Developed a CRM system utilized by 25+ internal users, reducing manual data workflows by **30%**.
- Engineered task and project management workflows within the client management system to improve team coordination.
- Designed REST APIs, optimized MongoDB database queries, and deployed applications to production.

Projects

ClientPro – Full-Stack CRM System | *Next.js, Supabase, PostgreSQL, NextAuth* — Live Demo | GitHub

- Spearheaded a multi-role CRM featuring role-based access control (RBAC) authentication using NextAuth.
- Designed a PostgreSQL database schema with Row Level Security (RLS) to enforce per-role data access securely.
- Built interactive dashboards, automated invoice generation, and streamlined email workflows.

SnapClass – AI Attendance System | *Python, Streamlit, Supabase, Computer Vision* — Landing Page | GitHub

- Built an AI-powered attendance system using facial recognition (Dlib) and voice biometrics (Resemblyzer), eliminating manual roll-calls.
- Engineered QR-based student enrollment with real-time PostgreSQL synchronization via Supabase.
- Deployed a dedicated Next.js landing page on Vercel and hosted the core Python application on Streamlit Cloud.

Product Management System | *React, Node.js, MySQL, Aiven* — GitHub

- Built a secure product management platform supporting full CRUD operations and real-time data synchronization.
- Designed a scalable backend architecture with Node.js and managed MySQL via Aiven for enterprise extensibility.

Achievements & Open Source

Hackathons & Competitions: Finalist at Paradigm Shift Hackathon (EmpowerHer platform); active participant on CodeChef and Codeforces.

Open Source Contributions: Active contributor to the JWOC (JGEC Winter of Code) program.

Campus Involvement: Volunteer at JGEC Coders Club; participant in the CFI robotics workshop.